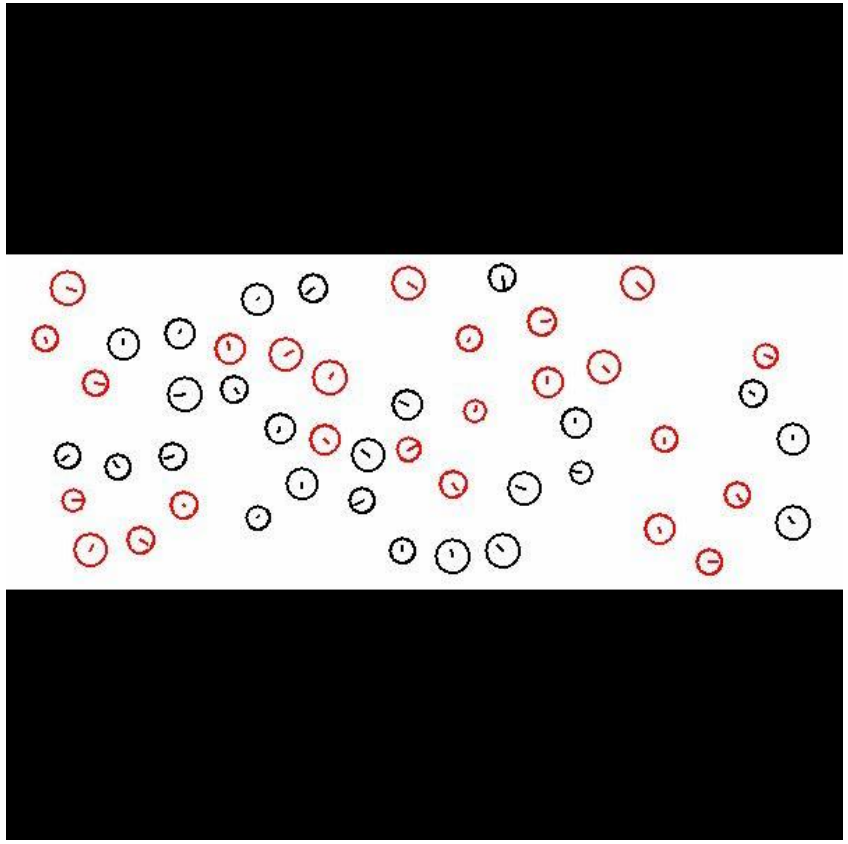
Data-driven modelling (3)

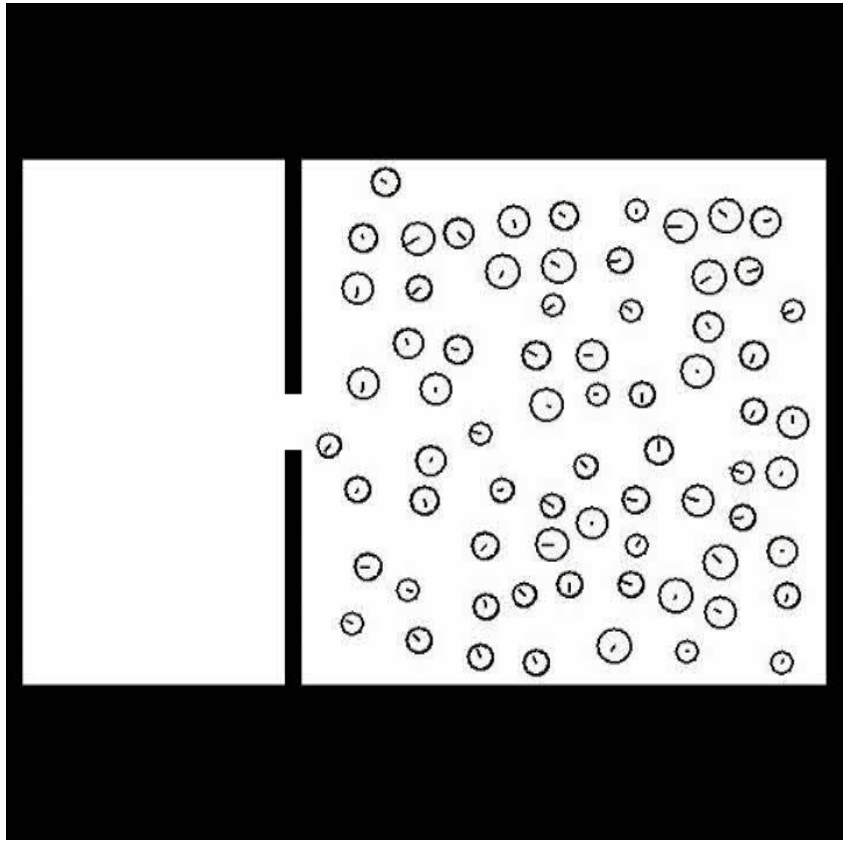
Microscopic and mesoscopic modelling

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Recap of microscopic models





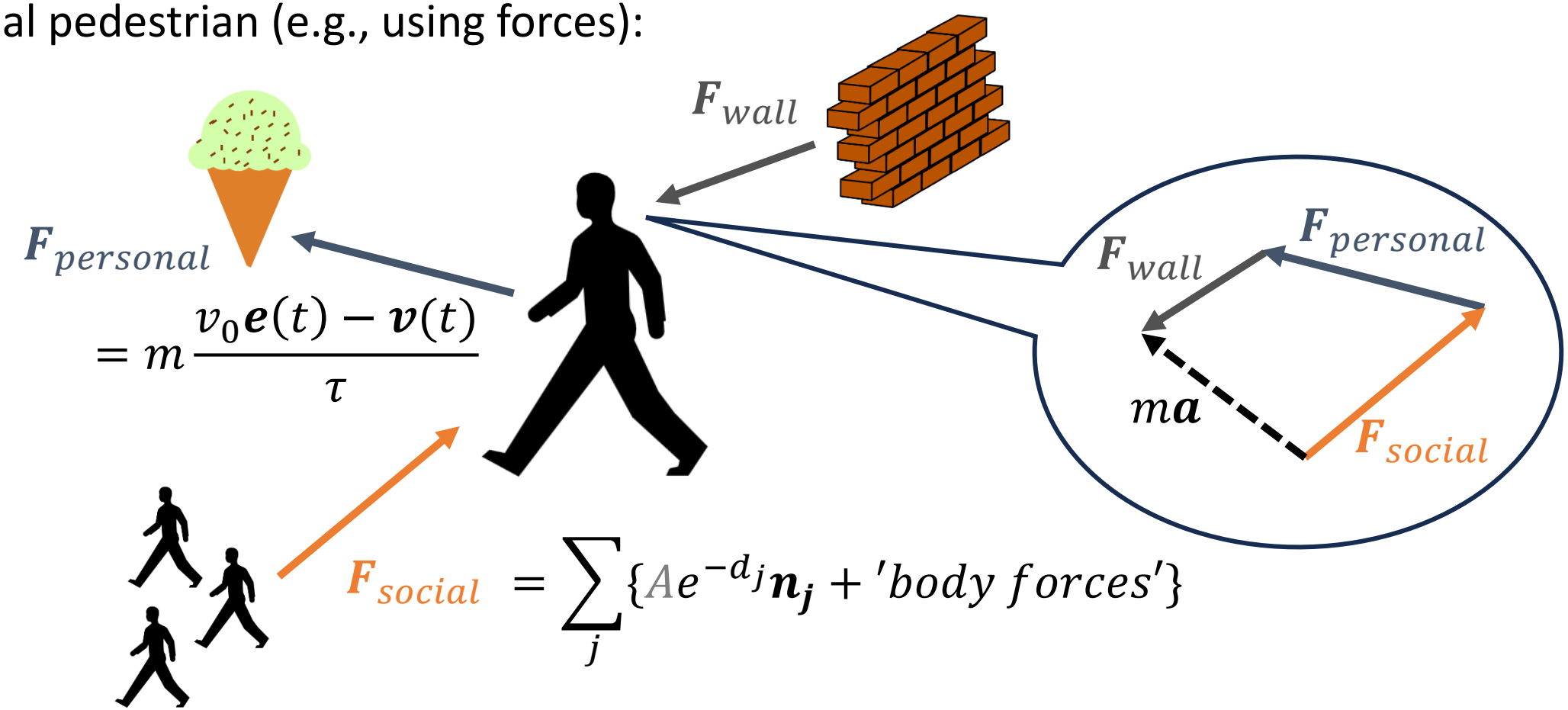
- Microscopic models describe the behaviour of individuals and are flexible modelling tools

Mechanistic microscopic models

Other model formulations are possible!

Often, we use maths to describe an individual pedestrian (e.g., using forces):

$$m\boldsymbol{a} = \boldsymbol{F}_{personal} + \boldsymbol{F}_{social} + \boldsymbol{F}_{wall}$$



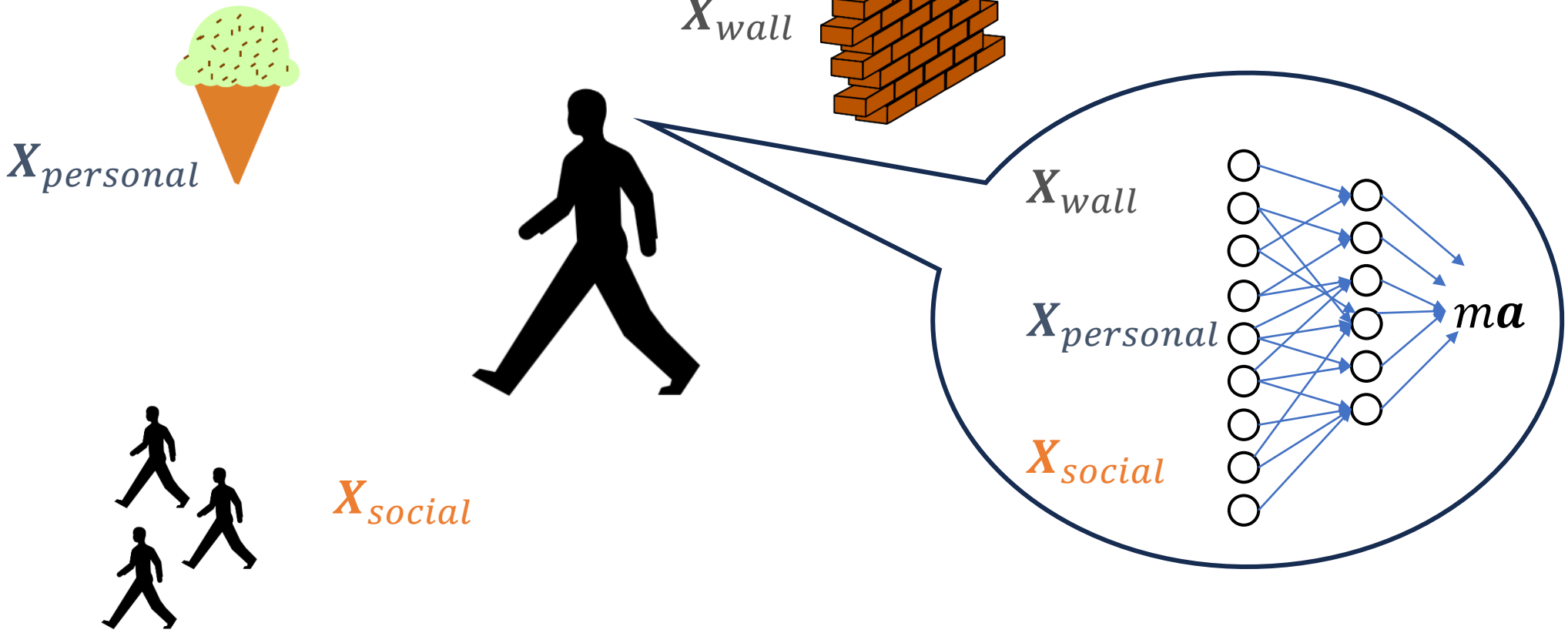
Mechanistic models - issues

- How do we decide on the mathematical algorithmic presentation of behaviour?
- How do we test the behaviour models assume is appropriate?
- Even mechanistic models have many parameters (m, v_0, τ, A on previous slide). How do we set their values?
- … all of this involves a lot of effort and ideally data, which leads to an alternative approach…

Learn behaviour of individuals from data

Attempt to learn the relationship between individuals' environment and their behaviour:

$$m\boldsymbol{a} = g(\boldsymbol{X}_{personal}, \boldsymbol{X}_{social}, \boldsymbol{X}_{wall}, \dots)$$

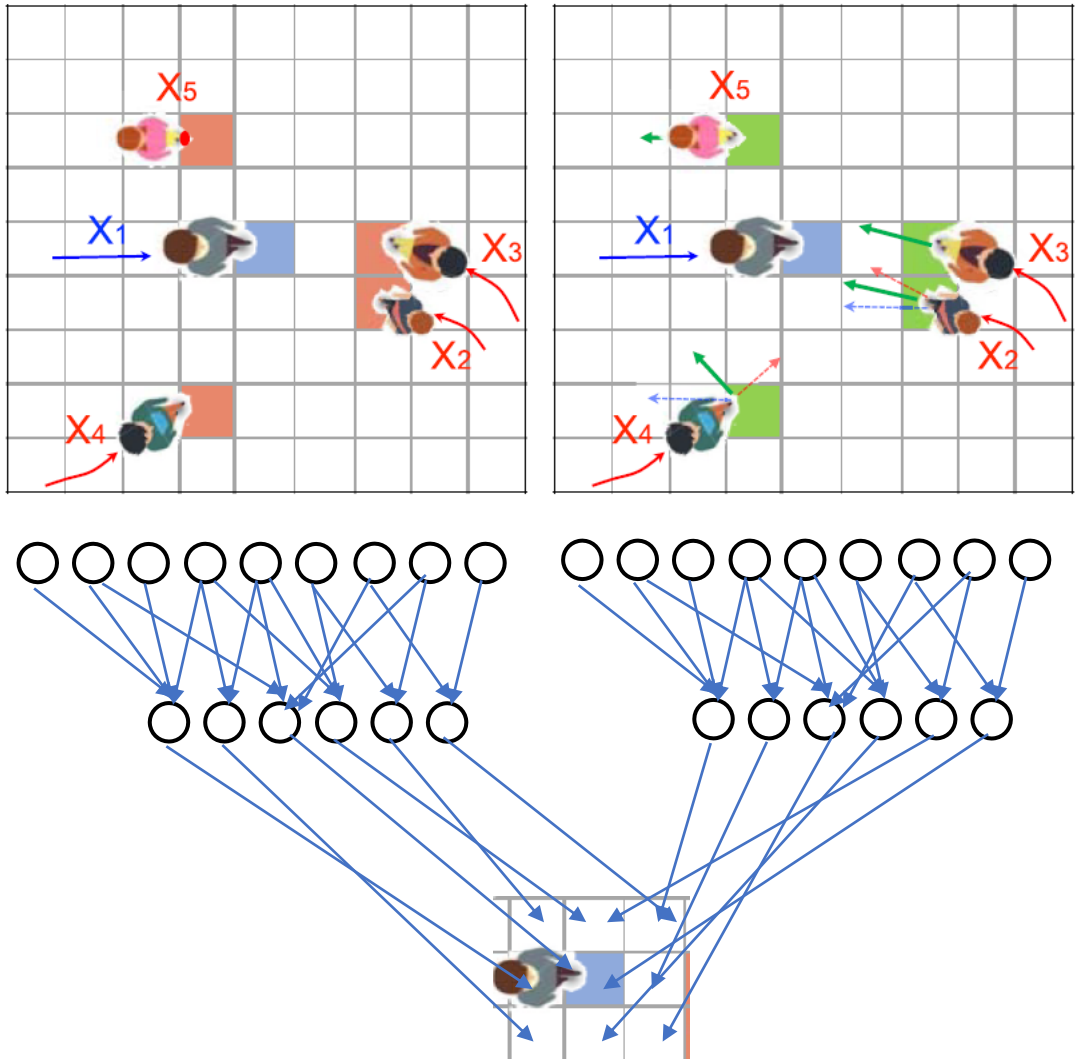


A simple example

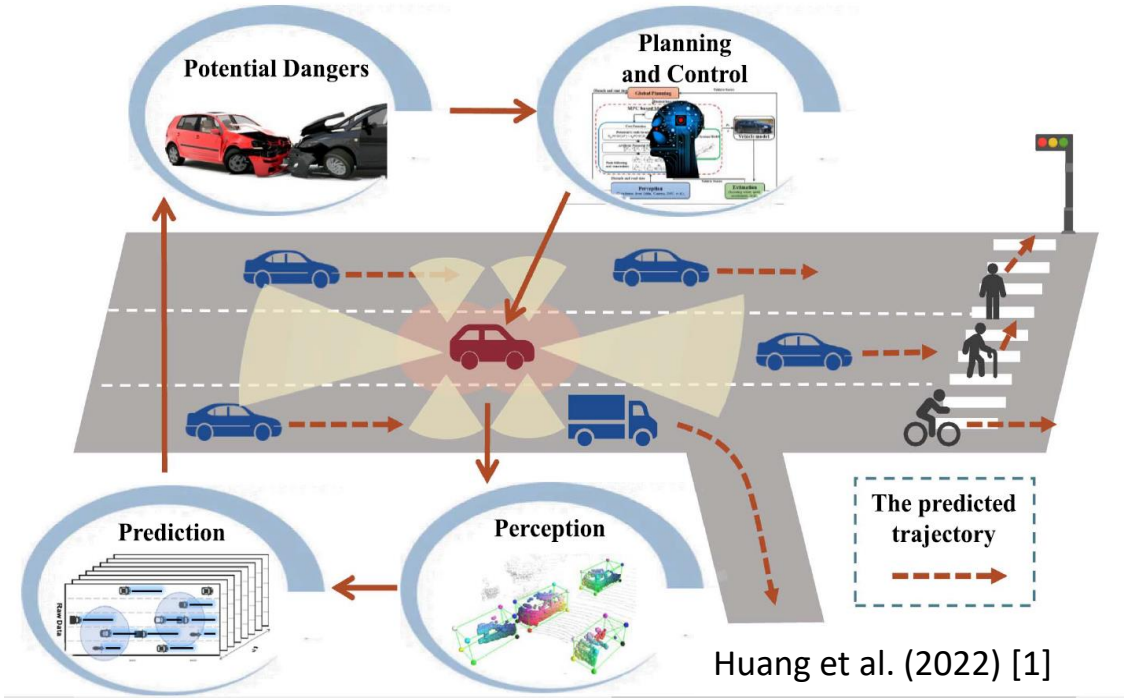
- Occupancy and movement directions of pedestrians around a focal pedestrian (blue cell) are captured using a grid. The grid is recentred whenever the focal pedestrian is changed.

- The resulting grid tensors are passed through an artificial neural network to predict the next position of the focal pedestrian.

Kothari et al. (2022) [2]



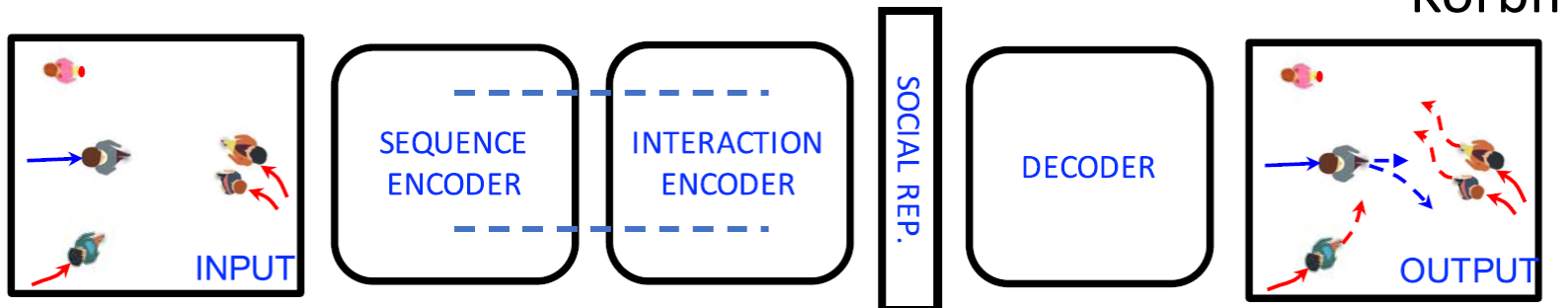
Examples



Huang et al. (2022) [1]

- Subject of extensive research (consider autonomous driving!).
- Examples for reviews:
  - Huang et al. (2022) [1]
  - Kothari et al. (2022) [2]
  - Korbmacher et al. (2022) [3]

Kothari et al. (2022) [2]



2D past coordinates of all pedestrians

Encodes past motion

Encodes neighbour interactions

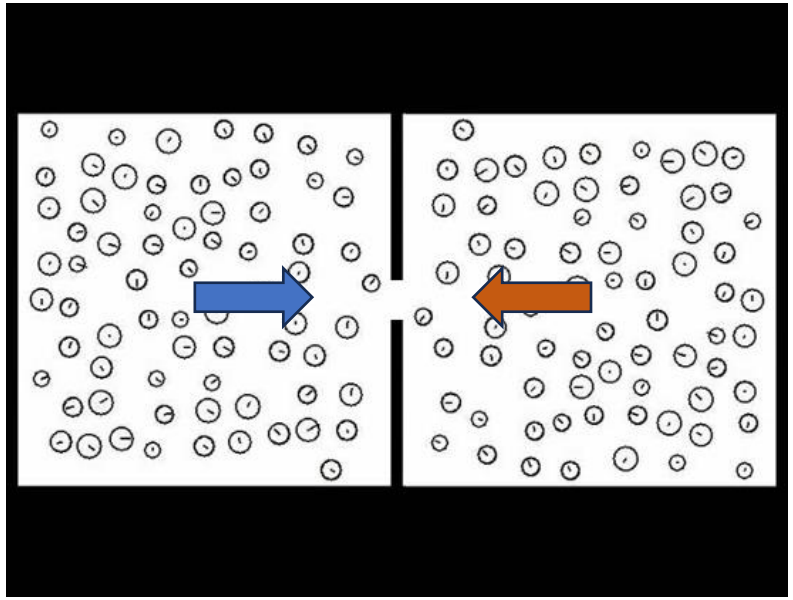
2D future coordinates of all pedestrians

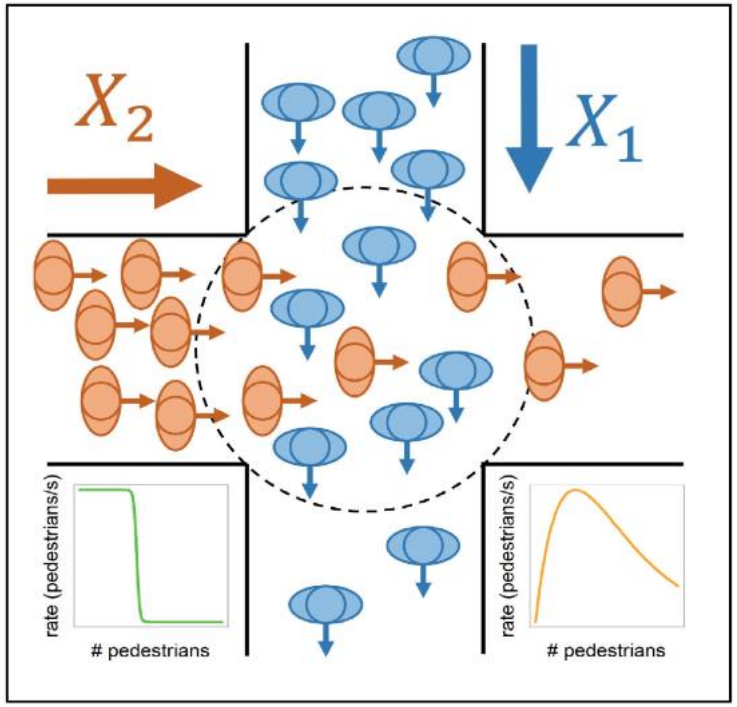
Learning microscopic models - issues

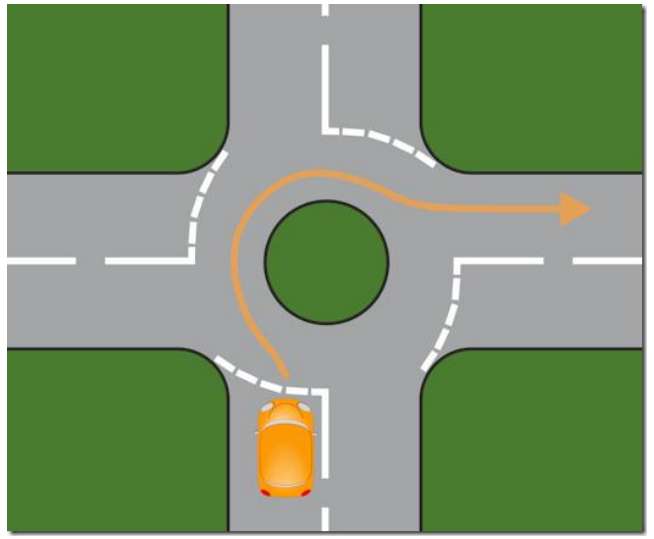
- Black box models – we do not necessarily improve our understanding of behaviour.
- Learned model may not extrapolate beyond the specific context of the data it was trained on.
- Some aspects of behaviour may be difficult/impossible to learn from data alone, e.g. long-term destinations.

Mesoscopic modelling approach

- Suppose we are interested in dynamics at a scale (time and space) somewhere between microscopic models and macroscopic models.







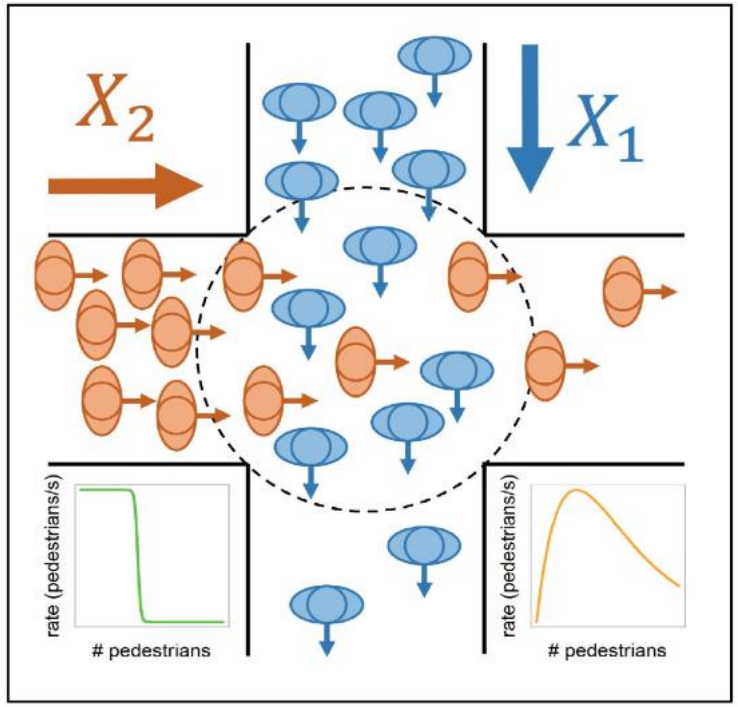
Learn model for dynamics at key infrastructure

- Let X be a variable describing the dynamics of the system we are interested in.
- Formulate a model:

$$\frac{\partial X}{\partial t} = f(X, Y, t)$$

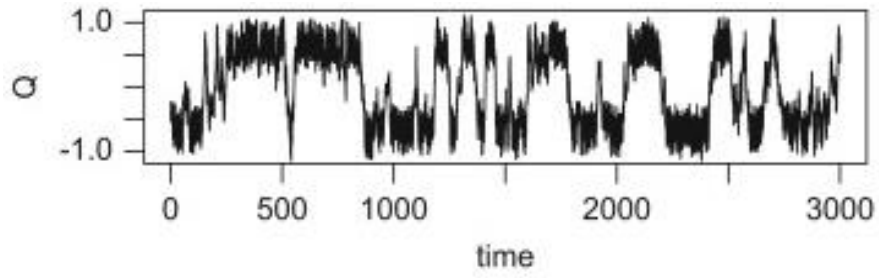
…where we don't know the functional form of f.

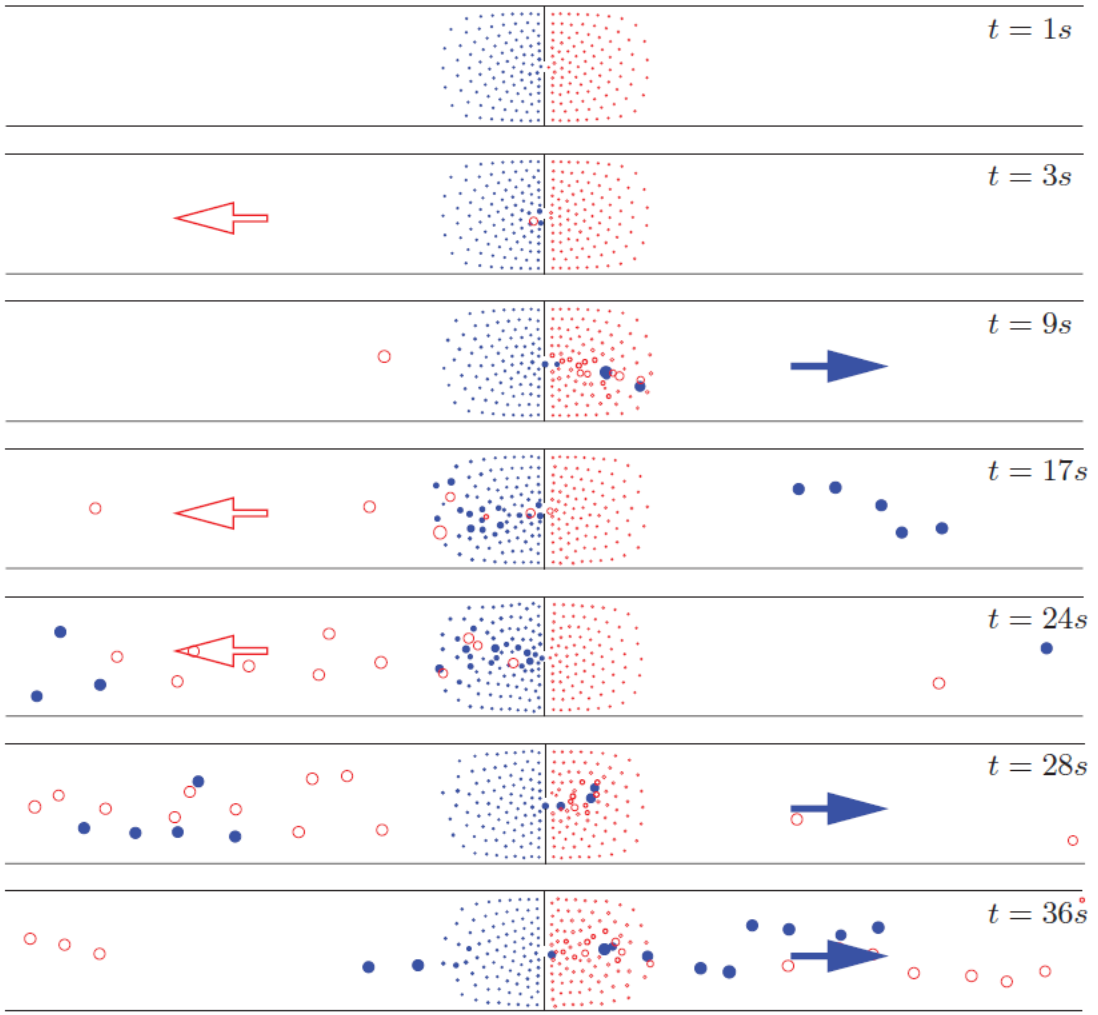
- Use data, or simulations of a microscopic model to approximate the shape of f.
- We need X to evolve at a different (slower) timescale than the microscopic dynamics. Choosing X appropriately is crucial!
- Notice the difference to black-box models.
- For an introduction reference [4] is useful, even though it is about animals. The following example is also based on [4].



Example (counter-flow at bottlenecks)

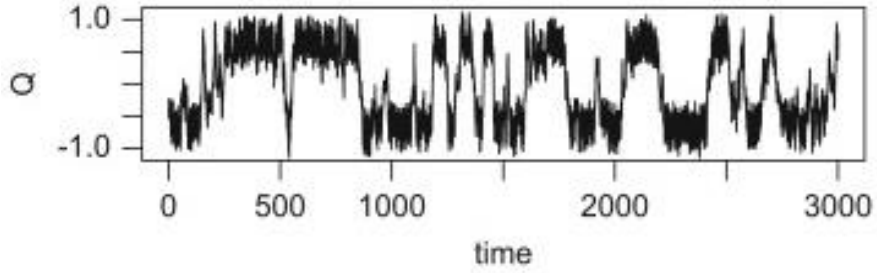
- Let flow to the left be positive, flow to right be negative, and let $Q \in [-1,1]$ be the normalised average flow through the bottleneck.
- For some bottleneck widths, we see:





Corradi et al. (2012)

Example (counter-flow at bottlenecks)



- Q captures the system well. It can be described using its time-dependent probability distribution $p(Q,t)$.
- We can hypothesise that $p(Q,t)$ can be approximately described using a Fokker-Planck equation:

$$\frac{\partial p}{\partial t} = \frac{\partial^2 (D(Q)p)}{\partial Q^2} + \frac{\partial (F(Q)p)}{\partial Q}$$

- The diffusion, $D(Q)$, and drift, $F(Q)$, capture the magnitude of randomness and mean rate of change in Q, respectively, and their functional forms need to be estimated.

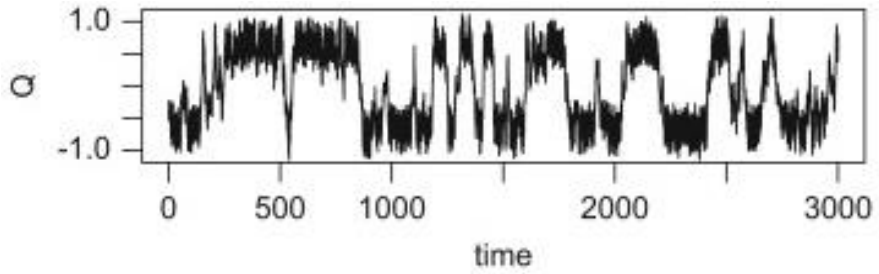
Example (counter-flow at bottlenecks)

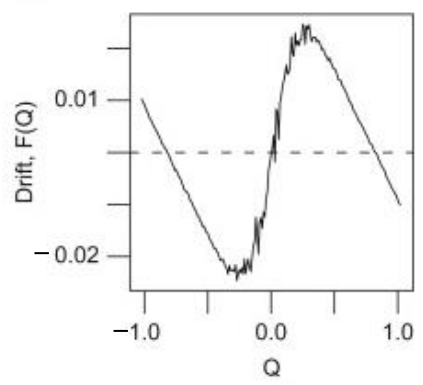
- We approximate

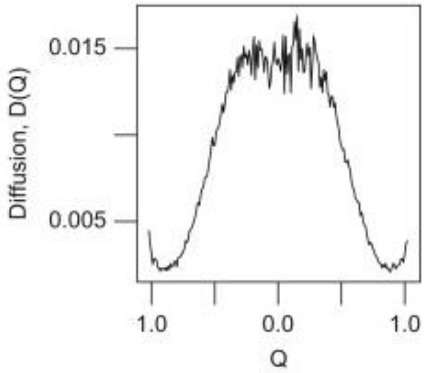
$$F(Q) \approx \left\langle \frac{Q(t+\delta t)-Q(t)}{\delta t} \right\rangle$$

$$D(Q) \approx \frac{1}{2}\left\langle \frac{(Q(t+\delta t)-Q(t))^2}{\delta t} \right\rangle$$

- For a small number of time steps δt in data (or simulations).
- In data, find all time points where Q takes the same value as the starting point for the equations above.
- Additional properties of the system can then be found, e.g. mean switching times.



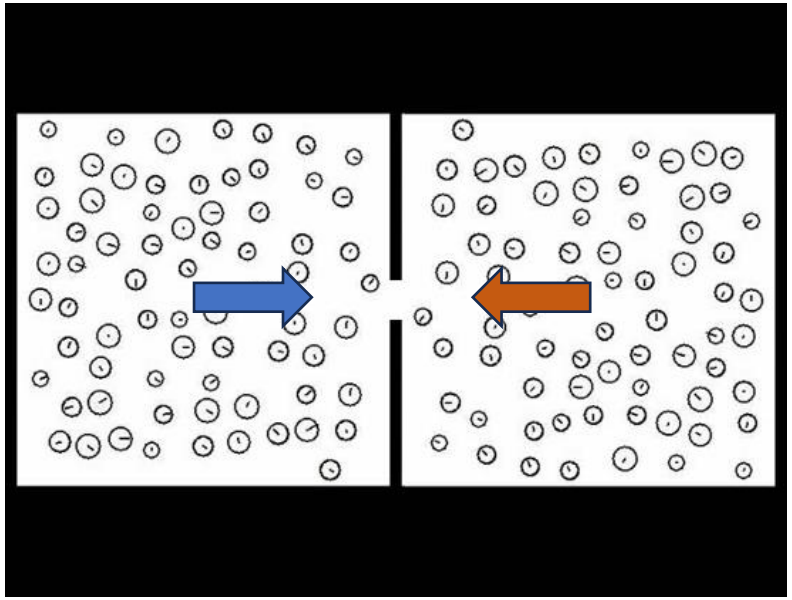




Equation-free modelling

recall: $\frac{\partial X}{\partial t} = f(X, Y, t)$

- When using simulations of a microscopic model instead of data, a commonly used technique is equation-free modelling.
- This involves four steps:
  - Lifting: creating microstates $x(t)$ consistent with macrostate $X(t)$.
  - Simulation: use microscopic simulations to compute $x(t)$ over a short time interval, δt.
  - Restriction: obtain $X(t + \delta t)$ from x$(t + \delta t)$.
  - Time-step: extrapolate the macrostate to $t + \Delta t$.



- Published examples (quite theoretical) :
  - Corradi, et al. (2012). Equation-free detection and continuation of a Hopf bifurcation point in a particle model of pedestrian flow. SIAM Journal on Applied Dynamical Systems, 11(3), 1007-1032.
  - Marschler, C., Sieber, J., Hjorth, P. G., & Starke, J. (2015). Equation-free analysis of macroscopic behavior in traffic and pedestrian flow. In Traffic and Granular Flow'13 (pp. 423-439). Springer International Publishing.

Summary

- Learning behaviour of individuals from data (microscopic models)
- Learning traffic dynamics at key infrastructure from data
  - Based on data
  - Based on simulations of microscopic models
- Equation-free modelling

Further reading

1. Huang, Y., et al. (2022). A survey on trajectory-prediction methods for autonomous driving. *IEEE Transactions on Intelligent Vehicles*, 7(3), 652-674.
2. Kothari, P., et al. (2021). Human trajectory forecasting in crowds: A deep learning perspective. *IEEE Transactions on Intelligent Transportation Systems*, 23(7), 7386-7400.
3. Korbmacher, R. et al. (2022). Review of pedestrian trajectory prediction methods: Comparing deep learning and knowledge-based approaches. *IEEE Transactions on Intelligent Transportation Systems*.
4. Yates, C. A., Erban, R., Escudero, C., Couzin, I. D., Buhl, J., Kevrekidis, I. G., ... & Sumpter, D. J. (2009). Inherent noise can facilitate coherence in collective swarm motion. Proceedings of the National Academy of Sciences, 106(14), 5464-5469.